

Meeting II: Intro to Category Theory Pt. 1

Topoi: The Categorical Analysis of Logic

Mohith Muddasani

May 19, 2026

Goals for today

By the end of this meeting, we should understand:

- ▶ the definition of a quiver, a path, and a category
- ▶ several examples of quivers and categories
- ▶ constructions of new categories from old ones

Quiver

Quiver: Definition

Definition

A **quiver** $Q = (V, E, s, t)$ comprises:

- ▶ a collection V of things called Q -vertices
- ▶ a collection E of things called Q -edges
- ▶ an operation $s : E \rightarrow V$ that maps each Q -edge e to a Q -object $s(e)$ (the “source”)
- ▶ an operation $t : E \rightarrow V$ that maps each Q -edge e to a Q -object $t(e)$ (the “target”)

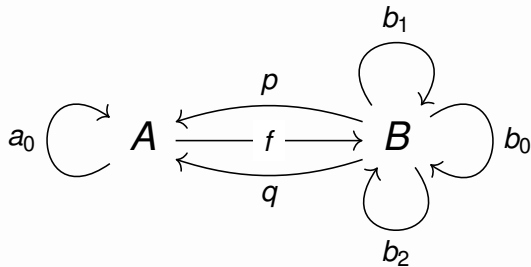
Notes

- ▶ A quiver is a directed graph that allows loops and multiple parallel edges.
- ▶ If V or E is a class (a.k.a. “large” set), we call the quiver “large”.
- ▶ If V and E are both sets (a.k.a. “small” sets), we call the quiver “small”.

Example: A Typical Quiver

$$A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B$$

Example: A Fancier Quiver



Example: An Arrowless Quiver

A B C

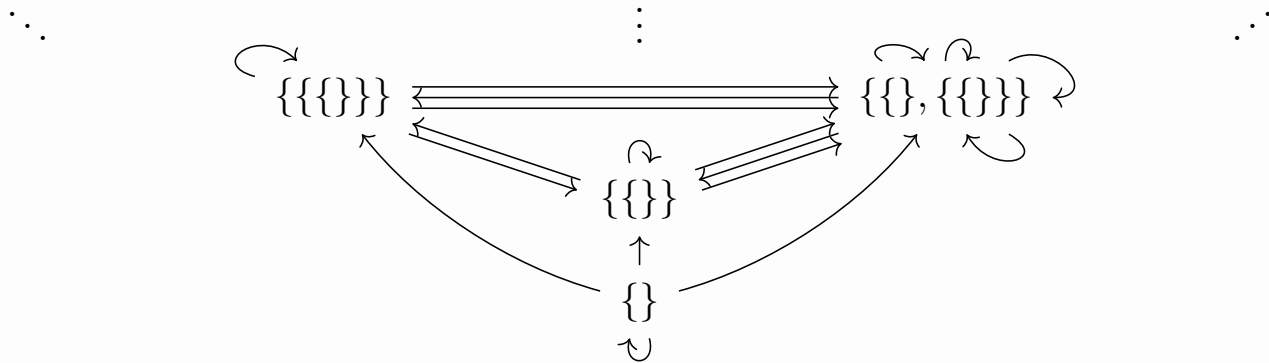
Example: The Empty Quiver

$$(\emptyset, \emptyset, !: \emptyset \rightarrow \emptyset, !: \emptyset \rightarrow \emptyset)$$

Intuition

- ▶ The above quiver is intentionally depicted as a blank picture.
- ▶ There is exactly one quiver with no vertices.

Example: A Large Quiver



Notes

- ▶ The hidden edge labels are meant to be distinct functions between sets.
- ▶ This quiver is **locally small**:
 - although the entire collection of edges is “large”,
 - each collection of edges between any pair of sets is “small”.

Path

Path: Definition

Definition

In a quiver $Q = (V, E, s, t)$, a **path** $P = (X, Y, \rho) \in V \times V \times \bigcup_{n \in \mathbb{N}} E^n$ from X to Y consists of a finite sequence of edges $\rho = \langle e_1, e_2, \dots, e_n \rangle$ such that:

- ▶ if there are no edges (length $n = 0$), then $X = Y$
- ▶ otherwise (length $n > 0$):
 - the first edge has source X
 - ▶ $X = s(e_1)$
 - consecutive pairs fit together in a head-to-tail fashion
 - ▶ $t(e_k) = s(e_{k+1})$ for $1 \leq k < n$
 - the last edge has target Y
 - ▶ $t(e_n) = Y$

Path: Definition

A path P can be depicted as a *diagram* P with the shape of a tree-like quiver that has $n + 1$ vertices and n edges:

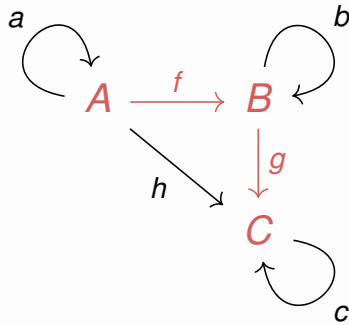
$$P : X = s(P) = s(e_1) \xrightarrow{e_1} t(e_1) = s(e_2) \xrightarrow{e_2} \cdots \xrightarrow{e_n} t(e_n) = t(P) = Y$$

Notes

In contrast to other conventions, for a quiver (V, E, s, t) , a quiver path $P = (X, Y, \langle e_1, \dots, e_n \rangle)$:

- ▶ has finite **length** n ; if $n = 0$, it is called an **empty** path
- ▶ has a “source” X (even when empty), denoted $s(P)$ by abuse of notation
- ▶ has a “target” Y (even when empty), denoted $t(P)$ by abuse of notation
- ▶ allows repetition of edges

Example: A Typical Path



$(A, C, \langle f, g \rangle)$

Path: Definition

Another way to think about a path diagram of length n in a quiver Q is as a *graph/quiver homomorphism* to Q from “the” *walking path quiver of length n* :

$$\begin{array}{ccccccc} 0 & \xrightarrow{\langle 0,1 \rangle} & 1 & \xrightarrow{\langle 1,2 \rangle} & \dots & \xrightarrow{\langle n-2,n-1 \rangle} & n-1 & \xrightarrow{\langle n-1,n \rangle} & n \\ & & & & \Downarrow & & & & \\ X_0 & \xrightarrow{e_1} & X_1 & \xrightarrow{e_2} & \dots & \xrightarrow{e_{n-1}} & X_{n-1} & \xrightarrow{e_n} & X_n \end{array}$$

Intuition

“the” walking singleton is $\{0\} \cong \{\mathbb{N}\}$

- ▶ in category theory, the article “the” refers to mathematical structure that is specified up-to-structure-preserving-isomorphism
- ▶ the adjective “walking” refers to “the” representative structure of a given kind

Example: A Path With Cyclic Image

$$A \xrightarrow{f} B \begin{array}{c} \xrightarrow{g} \\ \xleftarrow{s} \end{array} C \xrightarrow{h} D$$

Notes

As shown, the path is ambiguous. To fully specify the path, we must choose the sequence of edges. For example:

- ▶ $(B, B, \langle g, s \rangle)$
- ▶ $(C, C, \langle s, g \rangle)$
- ▶ $(B, C, \langle g, s, g, s, g \rangle)$
- ▶ $(C, B, \langle s, g, s \rangle)$

$$(B, B, \langle g, s, g, s \rangle)$$

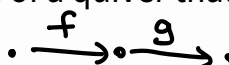
Example: An Empty Path

$$A \xrightarrow{f} B \begin{array}{c} \xrightarrow{g} \\ \xleftarrow{s} \end{array} C \xrightarrow{h} D$$

Path: Collections of Fixed Length

$$E_{t \times s} E := \{ \langle f, g \rangle \in E \times E : t(f) = s(g) \}$$

Given a quiver $Q = (V, E, s, t)$, we use the following abuses of notation conventions:

- ▶ Identify each $X \in V$ with the empty path $(X, X, \langle \rangle)$
 - Notice by composing abuses of notation conventions, $s(X) = X = t(X)$
- ▶ Identify each $e \in E$ with the singleton path $(s(e), t(e), \langle e \rangle)$
- ▶ Let $E_{t \times s} E \subseteq E \times E$ denote the collection of paths of length two:
 - Each $\langle f, g \rangle \in E_{t \times s} E$ can be depicted as a *diagram* with the shape of a quiver that has 3 vertices and 2 edges: $s(f) \xrightarrow{f} t(f) = s(g) \xrightarrow{g} t(g)$ 
- ▶ Let $E_{t \times s} E_{t \times s} E \subseteq E^3$ denote the collection of paths of length three

Intuition

Category theory is filled with convenient-to-use abuses of notation that are inconvenient-at-first to learn.

Category

Category: Definition

Definition

A **category** $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ comprises:

$$\cdot \xrightarrow{f} \cdot \xrightarrow{g} \cdot$$

- ▶ an underlying quiver $(O, A, \text{dom}, \text{cod})$
- ▶ a “composition” operation $\circ : A_{\text{cod} \times \text{dom}} A \rightarrow A$, denoted $\circ(\langle f, g \rangle) = g \circ f$, where:
 - Domain is preserved: $\text{dom}(g \circ f) = \text{dom}\langle f, g \rangle = \text{dom } f$
 - Codomain is preserved: $\text{cod}(g \circ f) = \text{cod}\langle f, g \rangle = \text{cod } g$
 - Associativity holds: $h \circ (g \circ f) = (h \circ g) \circ f$
- ▶ an “identity” operation $1 : O \rightarrow A$, denoted $1(X) = 1_X$, where:
 - $\text{dom } 1_X = X$
 - $\text{cod } 1_X = X$
 - $f \circ 1_{\text{dom } f} = f$
 - $1_{\text{cod } f} \circ f = f$

$$f: A \rightarrow B$$

$$f \circ 1_A = f \quad 1_B \circ f = f$$

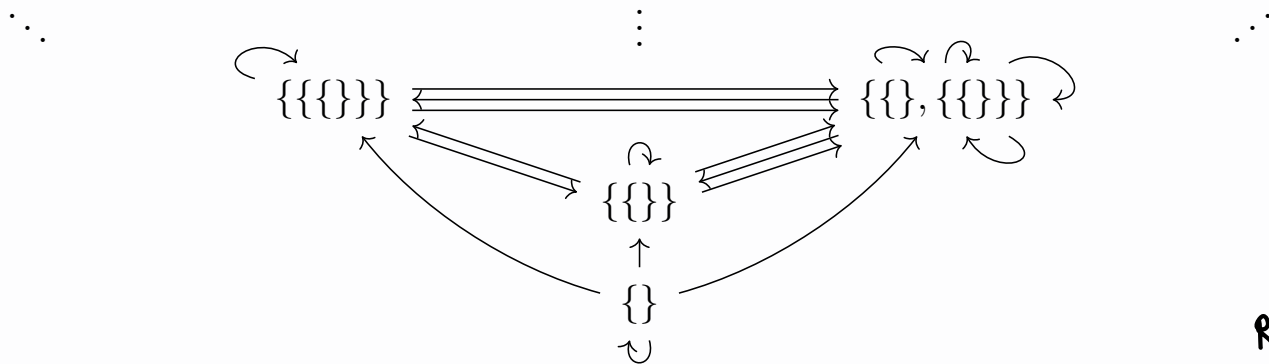
Category: Definition

Definition

Given a category $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$, we adopt the following conventions:

- ▶ The collection O of things is called $\mathcal{C}$ -**objects**, *vertices*
- ▶ The collection A of things is called $\mathcal{C}$ -**arrows**, *edges, morphisms, homomorphisms*
- ▶ The source operation $\text{dom} : A \rightarrow O$ is called **domain**
- ▶ The target operation $\text{cod} : A \rightarrow O$ is called **codomain**

Canonical Example: Set



Notes

- The underlying quiver is the same as earlier: objects are sets, arrows are functions
- Composition of $f : A \rightarrow B$ and $g : B \rightarrow C$ is the usual function composition:
 - $g \circ f = a \in A \mapsto g(f(a)) \in C$
- Each set X has a corresponding identity function: $1_X = x \in X \mapsto x \in X$

$$R \subseteq A \times B$$

$\langle A, B, R \rangle$ st. function

Category: Intuition

Intuition

► A category is a quiver that allows the reduction of finite paths to arrows.

- The empty path situated at X reduces to the arrow $X \xrightarrow{1_X} X$
- The path $X \xrightarrow{f} Y \xrightarrow{g} Z$ reduces to the **composite** arrow $X \xrightarrow{g \circ f} Z$

$A \xrightarrow{f} B$ "composes" to itself

$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$

$A \xrightarrow{f} B \xrightarrow{h \circ g} D$

$A \xrightarrow{g \circ f} C \xrightarrow{h} D$

$A \xrightarrow{(h \circ g) \circ f} D$

$A \xrightarrow{h \circ (g \circ f)} D$

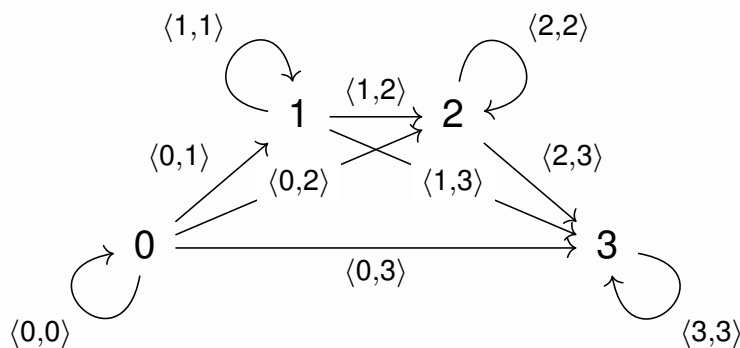
Examples of Categories

Example: The Empty Category

Intuition

- ▶ The above category is intentionally depicted as a blank picture.
- ▶ There is exactly one category with no objects.

Example: A Chain $0 \leq 1 \leq 2 \leq 3$



$$\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 10$$

Notes

- Identities and composites are uniquely determined.

Example: A Chain of Length n

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow n-1 \longrightarrow n$$

$$\binom{n+1}{2} + n+1$$

$$\langle k, k+1 \rangle$$

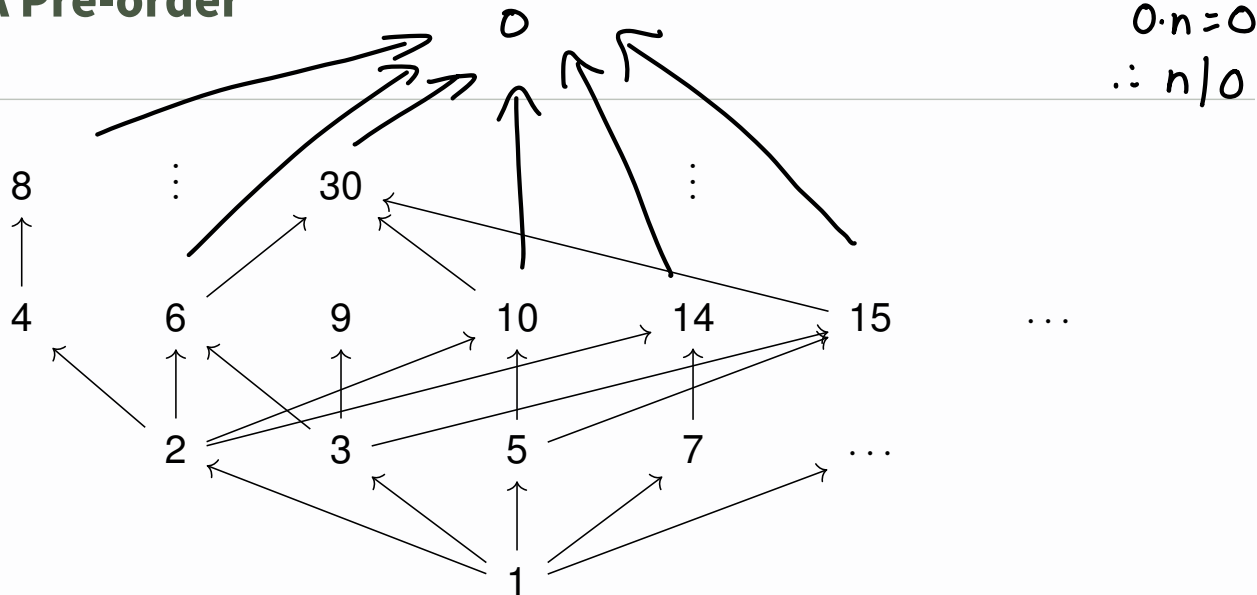
$$0 \leq k < n$$

are the
generating
morphisms

Notes

- Only generating arrows are being shown.
 - Identities and composite arrows have been left implicit.

Example: A Pre-order



Intuition

- This is (a fragment of) the divisibility pre-order on positive natural numbers.

Example: A Discrete Set/Class

$$S = \{1, 2, 3\}$$

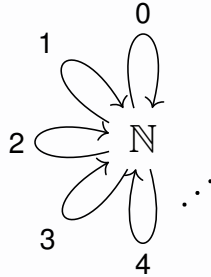
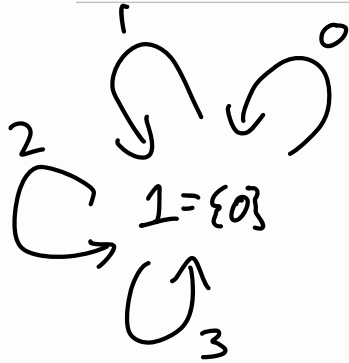


$$(S, S, 1_S, 1_S, \emptyset, 1_S)$$

$$\emptyset: S_{\text{cod}} \times_{\text{dom}} S = \{ \langle f, g \rangle : 1_S(f) = 1_S(g) \} = \{ \langle f, f \rangle : f \in S \}$$
$$\cong S$$

Example: The Monoid $(\mathbb{N}, +, 0)$

$$\mathbb{N}, \times, \mathbb{N} = \left\{ \langle m, n \rangle : \underbrace{!(m) = !(n)}_{\text{always true}} \right\} = \mathbb{N} \times \mathbb{N}$$



$$(1, \mathbb{N}, ! : \mathbb{N} \rightarrow 1, ! : \mathbb{N} \rightarrow 1, +, c_0)$$

$$c_0 : 1 \rightarrow \mathbb{N}$$

$$c_0(0) = 0$$

Notes

- Formally, the category is $(\{\mathbb{N}\}, \mathbb{N}, !, !, +, c_0)$ where $! : \mathbb{N} \rightarrow \{\mathbb{N}\}$ denotes the unique function into a singleton codomain, and $c_0 : \{\mathbb{N}\} \rightarrow \mathbb{N}$ denotes the function that takes its sole input ($\mathbb{N}$) to the output 0.
- The symbol “!” is often reserved for when a function/arrow uniquely satisfies a structural constraint.

$$! : \emptyset \rightarrow \mathbb{N}$$

Example: Vect_k

Fix a field k .

Objects: k -vector spaces

Morphisms: linear transformations

e.g. $k = \mathbb{R}$

$$\mathbb{R}^2 \xrightarrow{\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}} \mathbb{R}^3$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

etc.

Example: Grp

Objects: groups

Morphisms: group homomorphisms

$$\text{parity}: S_7 \longrightarrow S_2$$

Example: Top

Objects: topological spaces

Morphisms: continuous maps

$$\mathbb{R} \longleftrightarrow \mathbb{C}$$

Example: Mon

Objects: Monoids

Morphisms: monoid homomorphisms

$$(\mathbb{N}, +, 0) \hookrightarrow (\mathbb{Q}, +, 0)$$

Example: Pos

Example: Met

Beware

The choice of objects does not always offer an obvious choice of morphisms.

Constructing New Categories

Construction: Opposite Category

Definition

Let $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ be a category. Its **opposite category**, denoted $\mathcal{C}^{\text{op}}$, is $(O, A, \text{cod}, \text{dom}, \circ', 1)$ where $f \circ' g = g \circ f$.

Construction: Subcategory

Definition

Let $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ be a category. Let $O' \subseteq O$ and $A' \subseteq A$ be subsets closed under each of the operations, in the sense that:

- ▶ $1(O') \subseteq A'$
- ▶ $\text{dom}(A') \subseteq O'$
- ▶ $\text{cod}(A') \subseteq O'$
- ▶ $\circ(A'_{\text{cod} \times \text{dom}} A') \subseteq A'$

Then $\mathcal{C}' = (O', A', \text{dom}|_{A'}, \text{cod}|_{A'}, \circ|_{A'_{\text{cod} \times \text{dom}} A'}, 1|_{O'})$ is a subcategory of $\mathcal{C}$.

Guided Exercises

Exercise: Subcategories of a Monoid

Consider the natural numbers monoid as the category $(\{\mathbb{N}\}, \mathbb{N}, !_{\mathbb{N} \rightarrow \{\mathbb{N}\}}, !_{\mathbb{N} \rightarrow \{\mathbb{N}\}}, +, \mathbf{c}_0)$.
Let's explore its subcategories.

Exercise: Constructing the Sum Category

Exercise: Categories with Isomorphic Quivers

Exercise: Constructing the Product Category

Exercise: Constructing the Arrow Category

Exercise: Constructing the Slice Category

Beware

The book calls it a comma category, but this term is usually reserved for a more general construction that requires the prerequisite concept of a morphism of categories (called “functor”).

Outro

Reading for Next Lecture

- ▶ Minimal
 - Sections 1-8, 13 of chapter 3
- ▶ Excellent
 - All of chapter 3
- ▶ Extra
 - Limits and Colimits